MATH 152 — Part 5

Conic Sections

Detailed Notes with Tips and Examples

Contents

5.1 Introduction to Conic Sections

What are Conic Sections?

A **conic section** is a curve obtained by intersecting a **right circular cone** with a plane. The angle at which the plane cuts the cone determines the type of curve.

The Three Basic Conic Sections

Conic	Eccentricity (ε)	Description
Parabola	$\varepsilon = 1$	Distance to focus = Distance to directrix
Ellipse	$0 < \varepsilon < 1$	Distance to focus < Distance to directrix
Hyperbola	$\varepsilon > 1$	Distance to focus > Distance to directrix

Special Case: When $\varepsilon = 0$, the ellipse becomes a **circle**.

5.2 The Parabola

Definition

A **parabola** is the locus of a point P(x,y) such that its **distance from a fixed point** (called the **focus**) equals its **distance from a fixed line** (called the **directrix**).

Standard Equation of a Parabola

Vertex at origin, focus at (a, 0), directrix x = −a:

$$y^2 = 4ax$$

Derivation:

Let P(x, y) be any point on the parabola.

Distance from P to focus (a, 0):

$$PA = \sqrt{(x - a)^2 + (y - 0)^2} = \sqrt{(x - a)^2 + y^2}$$

Distance from P to directrix x = −a:

$$PM = |x + a|$$

By definition: PA = PM

$$\sqrt{(x - a)^2 + y^2} = x + a$$
Square both sides:
$$(x - a)^2 + y^2 = (x + a)^2$$
$$x^2 - 2ax + a^2 + y^2 = x^2 + 2ax + a^2$$
$$y^2 = 4ax$$

Key Features of $y^2 = 4ax$

Feature	Value
Vertex	(0, 0)
Focus	(a, 0)
Directrix	$x = -a$
Axis of symmetry	x-axis (y = 0)
Length of latus rectum	4a

Note: The parabola opens to the **right** for $a > 0$.

Other Forms of Parabola

Equation	Vertex	Focus	Directrix	Opens
$y^2 = 4ax$	(0,0)	(a,0)	$x = -a$	Right
$y^2 = -4ax$	(0,0)	(−a,0)	$x = a$	Left

Equation	Vertex	Focus	Directrix	Opens
$x^2 = 4ay$	(0,0)	(0,a)	$y = -a$	Up
$x^2 = -4ay$	(0,0)	(0,−a)	$y = a$	Down

Parametric Equations of Parabola

$x = at^2, \quad y = 2at$

Verification:

Substitute into $y^2 = 4ax$:

$$y^2 = (2at)^2 = 4a^2t^2$$
$$4ax = 4a(at^2) = 4a^2t^2$$
$$y^2 = 4ax \checkmark$$

The point $(at^2, 2at)$ is called the point 't' on the parabola.

Tangent to the Parabola

Gradient at any point:

From $y^2 = 4ax$:

$$2y \cdot dy/dx = 4a$$
$$dy/dx = 2a/y$$

At point 't' $(at^2, 2at)$:

$$dy/dx = 2a/(2at) = 1/t$$

Equation of tangent at point 't':

$$y - 2at = (1/t)(x - at^2)$$
$$ty - 2at^2 = x - at^2$$
$$ty - x - at^2 = 0$$

Equation of tangent at (x_1, y_1):

$$yy_1 = 2a(x + x_1)$$

Normal to the Parabola

Gradient of normal:

Since normal is perpendicular to tangent:

$$m_{normal} = -t$$

Equation of normal at point 't':

$$y - 2at = -t(x - at^2)$$
$$y - 2at = -tx + at^3$$
$$y + tx - 2at - at^3 = 0$$

Latus Rectum

The **latus rectum** is a chord passing through the focus and perpendicular to the axis.

For $y^2 = 4ax$, when $x = a$:

$$y^2 = 4a(a) = 4a^2$$
$$y = \pm 2a$$

Length of latus rectum = 4a

Example 1: Parabola Basics

Problem: Find the focus, directrix, and length of latus rectum for $y^2 = 12x$.

Solution:

Compare $y^2 = 12x$ with $y^2 = 4ax$:

$$4a = 12 \rightarrow a = 3$$

- **Focus:** $(a, 0) = (3, 0)$
- **Directrix:** $x = -a = -3$
- **Length of latus rectum:** $4a = 12$

Example 2: Finding Intersection Points

Problem: A line passing through the focus of $y^2 = 4ax$ with gradient 1 intersects the parabola at two points. Find these points.

Solution:

Focus is at $(a, 0)$.

Line through $(a, 0)$ with gradient 1:

$$y - 0 = 1(x - a)$$
$$y = x - a$$

Substitute into $y^2 = 4ax$:

$$(x - a)^2 = 4ax$$
$$x^2 - 2ax + a^2 = 4ax$$
$$x^2 - 6ax + a^2 = 0$$
$$x = [6a \pm \sqrt{(36a^2 - 4a^2)}]/2$$
$$x = [6a \pm \sqrt{(32a^2)}]/2$$
$$x = [6a \pm 4a\sqrt{2}]/2$$
$$x = 3a \pm 2a\sqrt{2}$$

When $x = 3a + 2a\sqrt{2}$:
$$y = x - a = 2a + 2a\sqrt{2} = 2a(1 + \sqrt{2})$$

When $x = 3a - 2a\sqrt{2}$:
$$y = x - a = 2a - 2a\sqrt{2} = 2a(1 - \sqrt{2})$$

Points: $(3a \pm 2a\sqrt{2}, 2a(1 \pm \sqrt{2}))$

Example 3: Intersection of Normal with Parabola

Problem: The normal to $y^2 = 4ax$ at point t ($t \neq 0$) meets the curve again at point t'. Find t' in terms of t.

Solution:

Normal at t: $y + tx - 2at - at^3 = 0$

This line intersects the parabola $y^2 = 4ax$ again at t'.

At point t': $x = at'^2$, $y = 2at'$

Substitute into normal equation:

```
2at' + t(at'²) - 2at - at³ = 0
a(2t' + tt'² - 2t - t³) = 0
2t' + tt'² - 2t - t³ = 0
tt'² + 2t' - (2t + t³) = 0
```

This is a quadratic in t'. One root is t (the original point).

The sum of roots $= -2/t$

So: $t + t' = -2/t$

```
t' = -2/t - t = -(t² + 2)/t
```

Therefore: $t' = -(t^2 + 2)/t$

5.3 The Ellipse

Definition

An **ellipse** is the locus of a point P(x,y) such that its distance from a fixed point (the **focus**) is **always less than** its distance from a fixed line (the **directrix**).

The ratio of these distances is called the **eccentricity** ε, where $0 < \varepsilon < 1$.

Standard Equation of an Ellipse

Center at origin, foci on x-axis:

```
x²/a² + y²/b² = 1
```

where:

```
b² = a²(1 - ε²)
```

Key Relationships:

- **Semi-major axis:** a (larger)
- **Semi-minor axis:** b (smaller)

- **Foci:** $(\pm a\varepsilon, 0) = (\pm c, 0)$ where $c = a\varepsilon$
- **Directrices:** $x = \pm a/\varepsilon$
- **Eccentricity:** $\varepsilon = c/a$ $(0 < \varepsilon < 1)$

Note: Since $0 < \varepsilon < 1$, we have $a > b$.

Key Features of $x^2/a^2 + y^2/b^2 = 1$

Feature	Value
Center	$(0, 0)$
Vertices	$(\pm a, 0)$
Co-vertices	$(0, \pm b)$
Foci	$(\pm c, 0)$ where $c^2 = a^2 - b^2$
Length of major axis	$2a$
Length of minor axis	$2b$

Parametric Equations of Ellipse

$$x = a \cos t, \quad y = b \sin t$$

Verification:

Substitute into $x^2/a^2 + y^2/b^2 = 1$:

$$(a^2\cos^2 t)/a^2 + (b^2\sin^2 t)/b^2 = \cos^2 t + \sin^2 t = 1 \checkmark$$

The point (a cos t, b sin t) is called the point ‘t’ on the ellipse.

Tangent to the Ellipse

Gradient:

From $x^2/a^2 + y^2/b^2 = 1$:

$$2x/a^2 + 2y/b^2 \cdot dy/dx = 0$$
$$dy/dx = -b^2x/(a^2y)$$

At point (x_1, y_1):

$$dy/dx = -b^2x_1/(a^2y_1)$$

Equation of tangent at (x_1, y_1):

$$xx_1/a^2 + yy_1/b^2 = 1$$

Equation of tangent at point ‘t’ (a cos t, b sin t):

$$x \cos t/a + y \sin t/b = 1$$

Normal to the Ellipse

Gradient of normal:

$$m\_normal = a^2y_1/(b^2x_1)$$

Equation of normal at (x_1, y_1):

$$a^2y_1(x - x_1) - b^2x_1(y - y_1) = 0$$

Or in simplified form:

$$a^2y_1x - b^2x_1y = (a^2 - b^2)x_1y_1$$

Latus Rectum of Ellipse

When $x = c$ (at the focus):

$$c^2/a^2 + y^2/b^2 = 1$$
$$y^2/b^2 = 1 - c^2/a^2 = (a^2 - c^2)/a^2 = b^2/a^2$$
$$y = \pm b^2/a$$

Length of latus rectum = $2b^2/a$

Example 1: Ellipse Basics

Problem: Find the foci, vertices, and eccentricity of $x^2/25 + y^2/9 = 1$.

Solution:

$$a^2 = 25 \rightarrow a = 5$$
$$b^2 = 9 \rightarrow b = 3$$

$$c^2 = a^2 - b^2 = 25 - 9 = 16 \rightarrow c = 4$$

Vertices: $(\pm 5, 0)$
Co-vertices: $(0, \pm 3)$
Foci: $(\pm 4, 0)$
Eccentricity: $\varepsilon = c/a = 4/5 = 0.8$

Example 2: Finding Equation from Given Information

Problem: Find the equation of the ellipse with foci at $(\pm 4, 0)$ and major axis length 10.

Solution:

Major axis length = $2a = 10 \rightarrow a = 5$
Foci at $(\pm c, 0) = (\pm 4, 0) \rightarrow c = 4$

$$b^2 = a^2 - c^2 = 25 - 16 = 9 \rightarrow b = 3$$

Equation: $x^2/25 + y^2/9 = 1$

Example 3: Tangent to Ellipse

Problem: Find the equations of the tangents to $x^2/25 + y^2/9 = 1$ where the latus rectum meets the curve.

Solution:

Latus rectum: $x = c = 4$

When $x = 4$:

$$16/25 + y^2/9 = 1$$
$$y^2/9 = 9/25$$
$$y^2 = 81/25$$
$$y = \pm 9/5$$

Points on latus rectum: $(4, \pm 9/5)$

Tangent at (x_1, y_1): $xx_1/25 + yy_1/9 = 1$

At $(4, 9/5)$:

$$4x/25 + (9y/5)/9 = 1$$
$$4x/25 + y/5 = 1$$
$$4x + 5y = 25$$

At $(4, -9/5)$:

$$4x/25 - y/5 = 1$$
$$4x - 5y = 25$$

5.4 The Hyperbola

Definition

A **hyperbola** is the locus of a point $P(x,y)$ such that its distance from a fixed point (the **focus**) is **always greater than** its distance from a fixed line (the **directrix**).

The ratio of these distances is called the **eccentricity** ε, where $\varepsilon > 1$.

Standard Equation of a Hyperbola

Center at origin, foci on x-axis:

$$x^2/a^2 - y^2/b^2 = 1$$

where:

$$b^2 = a^2(\varepsilon^2 - 1)$$

Key Relationships:

- **Real axis (transverse axis):** $2a$ (along x-axis)
- **Imaginary axis (conjugate axis):** $2b$ (along y-axis)
- **Foci:** $(\pm a\varepsilon, 0) = (\pm c, 0)$ where $c = a\varepsilon$

- **Directrices:** $x = \pm a/\varepsilon$
- **Eccentricity:** $\varepsilon = c/a$ ($\varepsilon > 1$)

Note: Since $\varepsilon > 1$, we have $c > a$.

Key Features of $x^2/a^2 - y^2/b^2 = 1$

Feature	Value
Center	(0, 0)
Vertices	$(\pm a, 0)$
Foci	$(\pm c, 0)$ where $c^2 = a^2 + b^2$
Transverse axis length	2a
Conjugate axis length	2b

Asymptotes of Hyperbola

The asymptotes are lines that the hyperbola approaches as $|x| \to \infty$.

For $x^2/a^2 - y^2/b^2 = 1$, the asymptotes are:

$$y = \pm(b/a)x$$

Verification:

As $x \to \infty$, $x^2/a^2 \to \infty$, so $y^2/b^2 \approx x^2/a^2 - 1 \approx x^2/a^2$

$y \approx \pm(b/a)x$

Key Insight: The asymptotes are tangents to the hyperbola at infinity.

Parametric Equations of Hyperbola

Using sec and tan:

$$x = a \sec t, \quad y = b \tan t$$

Using cosh and sinh:

$$x = a \cosh t, \quad y = b \sinh t$$

Verification (using sec/tan):

$$x^2/a^2 - y^2/b^2 = \sec^2 t - \tan^2 t = 1 \checkmark$$

The Rectangular Hyperbola

Definition: A hyperbola where the asymptotes are perpendicular.

Condition: $a = b$

Equation:

$$x^2 - y^2 = a^2$$

Special Form ($c^2 = xy$):

When rotated 45°, the rectangular hyperbola takes the form:

$$xy = c^2$$

Parametric Equations:

$$x = ct, \quad y = c/t$$

Verification:

$$xy = (ct)(c/t) = c^2 \checkmark$$

Tangent to the Hyperbola

At point (x_1, y_1):

$$xx_1/a^2 - yy_1/b^2 = 1$$

At point 't' (a sec t, b tan t):

$$x \sec t/a - y \tan t/b = 1$$

Normal to the Hyperbola

At point (x_1, y_1):

$$a^2y_1(x - x_1) + b^2x_1(y - y_1) = 0$$

Latus Rectum of Hyperbola

When x = c (at the focus):

$$c^2/a^2 - y^2/b^2 = 1$$
$$y^2/b^2 = c^2/a^2 - 1 = (c^2 - a^2)/a^2 = b^2/a^2$$
$$y = \pm b^2/a$$

Length of latus rectum = $2b^2/a$

Example 1: Hyperbola Basics

Problem: Find the foci, vertices, asymptotes, and eccentricity of $x^2/16 - y^2/9 = 1$.

Solution:

$$a^2 = 16 \rightarrow a = 4$$
$$b^2 = 9 \rightarrow b = 3$$

$$c^2 = a^2 + b^2 = 16 + 9 = 25 \rightarrow c = 5$$

Vertices: $(\pm 4, 0)$
Foci: $(\pm 5, 0)$
Asymptotes: $y = \pm(3/4)x$

Eccentricity: $\varepsilon = c/a = 5/4 = 1.25$

Example 2: Rectangular Hyperbola

Problem: Find the equation of the tangent and normal to $xy = c^2$ at point t.

Solution:

For $xy = c^2$ with $x = ct$, $y = c/t$:

Tangent:

From $xy = c^2$:
$x \cdot dy/dx + y = 0$
$dy/dx = -y/x = -(c/t)/(ct) = -1/t^2$

At point (ct, c/t):

$y - c/t = (-1/t^2)(x - ct)$
$t^2 y - ct = -x + ct$
$x + t^2 y = 2ct$

Normal:

Gradient $= t^2$
$y - c/t = t^2(x - ct)$
$y - c/t = t^2 x - ct^3$
$y - t^2 x + ct^3 - c/t = 0$

Example 3: Intersection of Normal with Rectangular Hyperbola

Problem: The normal to $xy = c^2$ at point (ct, c/t) meets the hyperbola again at point t′. Find t′ in terms of t.

Solution:

Normal equation: $y - t^2 x + ct^3 - c/t = 0$

At point t′: $x = ct'$, $y = c/t'$

Substitute:

$c/t' - t^2(ct') + ct^3 - c/t = 0$
Divide by c:
$1/t' - t^2 t' + t^3 - 1/t = 0$

Multiply by t′:

$1 - t^2 t'^2 + t^3 t' - t'/t = 0$
$t^2 t'^2 - t^3 t' + t'/t - 1 = 0$

One root is $t' = t$. Factor:

$(t' - t)(?) = 0$

The other root: $t' = -1/t^3$

Therefore: $t' = -1/t^3$

5.5 Comparison of Conic Sections

Summary Table

Feature	Parabola	Ellipse	Hyperbola
Eccentricity ε	$\varepsilon = 1$	$0 < \varepsilon < 1$	$\varepsilon > 1$
Standard Equation	$y^2 = 4ax$	$x^2/a^2 + y^2/b^2 = 1$	$x^2/a^2 - y^2/b^2 = 1$
Relationship	—	$b^2 = a^2(1-\varepsilon^2)$	$b^2 = a^2(\varepsilon^2-1)$
Center	(0,0)	(0,0)	(0,0)
Vertices	(0,0)	$(\pm a, 0)$	$(\pm a, 0)$
Foci	(a,0)	$(\pm a\varepsilon, 0)$	$(\pm a\varepsilon, 0)$
Directrix	$x = -a$	$x = \pm a/\varepsilon$	$x = \pm a/\varepsilon$
Asymptotes	None	None	$y = \pm(b/a)x$
Parametric	$(at^2, 2at)$	$(a \cos t, b \sin t)$	$(a \sec t, b \tan t)$
Latus Rectum	4a	$2b^2/a$	$2b^2/a$

5.6 Standard Forms and Transformations

General Second-Degree Equation

The general equation of a conic is:

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

Identifying the Conic:

Condition	Type
$B = 0$ and $A = C$	Circle
$B = 0$ and $A \neq C$, same sign	Ellipse
$B = 0$ and A, C opposite signs	Hyperbola
Either $A = 0$ or $C = 0$	Parabola

Translations

To shift the conic to a new center (h, k):

Replace:

```
x → x - h
y → y - k
```

Examples:

Original	Shifted
$y^2 = 4ax$	$(y - k)^2 = 4a(x - h)$
$x^2/a^2 + y^2/b^2 = 1$	$(x-h)^2/a^2 + (y-k)^2/b^2 = 1$
$x^2/a^2 - y^2/b^2 = 1$	$(x-h)^2/a^2 - (y-k)^2/b^2 = 1$

Example: Shifting a Conic

Problem: Find the equation of the parabola with vertex at (2, 3) and focus at (5, 3).

Solution:

Focus is to the right of vertex → parabola opens right.

```
a = distance from vertex to focus = 5 - 2 = 3

Standard form with vertex at (h, k):
(y - k)² = 4a(x - h)
(y - 3)² = 12(x - 2)
```

Practice Problems and Solutions

Problem 1

Find the equation of the parabola with focus at (2, 0) and directrix x = −2.

Solution:

```
Focus: (2, 0) → a = 2
Directrix: x = -2 → x = -a ✓

Equation: y² = 4ax = 4(2)x = 8x
```

Problem 2

Find the vertex, focus, and directrix of $y^2 = -16x$.

Solution:

```
Compare with y² = -4ax:
-4a = -16 → a = 4

Vertex: (0, 0)
Focus: (-a, 0) = (-4, 0)
Directrix: x = a = 4
Opens: Left
```

Problem 3

Find the equation of the ellipse with vertices at (±5, 0) and foci at (±3, 0).

Solution:

$a = 5$ (distance from center to vertices)
$c = 3$ (distance from center to foci)

$b^2 = a^2 - c^2 = 25 - 9 = 16 \rightarrow b = 4$

Equation: $x^2/25 + y^2/16 = 1$

Problem 4

Find the equation of the hyperbola with vertices at (±4, 0) and foci at (±6, 0).

Solution:

$a = 4, c = 6$

$b^2 = c^2 - a^2 = 36 - 16 = 20$

Equation: $x^2/16 - y^2/20 = 1$
Asymptotes: $y = \pm(\sqrt{20}/4)x = \pm(\sqrt{5}/2)x$

Problem 5

Find the tangent to $y^2 = 8x$ at the point where $x = 2$.

Solution:

$y^2 = 8x \rightarrow 4a = 8 \rightarrow a = 2$

At $x = 2$: $y^2 = 16 \rightarrow y = \pm 4$

Points: (2, 4) and (2, -4)

Tangent at (x_1, y_1): $yy_1 = 2a(x + x_1)$

At (2, 4):
$4y = 4(x + 2) \rightarrow y = x + 2$

At (2, -4):
$-4y = 4(x + 2) \rightarrow y = -x - 2$

Problem 6

Find the tangent to $x^2/9 + y^2/4 = 1$ at the point where $x = 3/2$ (first quadrant).

Solution:

At $x = 3/2$:
$(9/4)/9 + y^2/4 = 1$
$1/4 + y^2/4 = 1$
$y^2/4 = 3/4$
$y^2 = 3$
$y = \sqrt{3}$ (first quadrant)

Point: $(3/2, \sqrt{3})$

$a^2 = 9$, $b^2 = 4$

Tangent: $xx_1/a^2 + yy_1/b^2 = 1$
$x(3/2)/9 + y(\sqrt{3})/4 = 1$
$x/6 + \sqrt{3}y/4 = 1$
$2x + 3\sqrt{3}y = 12$

Problem 7

Find the equation of the rectangular hyperbola whose asymptotes are x = 2 and y = 1, and which passes through (3, 2).

Solution:

Asymptotes: $x = 2$ and $y = 1$

The center is at (2, 1).

Rectangular hyperbola: $(x-2)(y-1) = c^2$

Passes through (3, 2):

$(3-2)(2-1) = 1 = c^2$
$c^2 = 1$

Equation: $(x-2)(y-1) = 1$

Problem 8

Find the length of the latus rectum of $x^2/16 - y^2/9 = 1$.

Solution:

$a^2 = 16 \rightarrow a = 4$
$b^2 = 9 \rightarrow b = 3$

Length of latus rectum $= 2b^2/a = 2(9)/4 = 18/4 = 4.5$

Problem 9

Find the eccentricity of the ellipse $x^2/16 + y^2/9 = 1$.

Solution:

$$a^2 = 16 \rightarrow a = 4$$
$$b^2 = 9 \rightarrow b = 3$$

$$c^2 = a^2 - b^2 = 16 - 9 = 7 \rightarrow c = \sqrt{7}$$

$$\varepsilon = c/a = \sqrt{7}/4 \approx 0.661$$

Problem 10

Find the directrix of $y^2 = 20x$.

Solution:

$$y^2 = 20x \rightarrow 4a = 20 \rightarrow a = 5$$

Directrix: $x = -a = -5$

Common Mistakes and How to Avoid Them

Mistake 1: Confusing a and b in Ellipse

Wrong:

$x^2/9 + y^2/16 = 1 \rightarrow a = 3, b = 4$ ✗

Correct:

$x^2/9 + y^2/16 = 1 \rightarrow a^2 = 9 \rightarrow a = 3, b^2 = 16 \rightarrow b = 4$
a is with x, b is with y

Mistake 2: Forgetting the Relationship $b^2 = a^2 - c^2$ (Ellipse)

Wrong:

$c^2 = a^2 + b^2$ ✗

Correct:

$c^2 = a^2 - b^2$ ✓

Mistake 3: Forgetting the Relationship $c^2 = a^2 + b^2$ (Hyperbola)

Wrong:

$c^2 = a^2 - b^2$ ✗

Correct:

$c^2 = a^2 + b^2$ ✓

Mistake 4: Confusing Focus and Vertex

Parabola:

- Vertex: (0, 0)
- Focus: (a, 0)
- They are NOT the same!

Mistake 5: Not Recognizing the Special Cases

- Circle is a special case of ellipse ($\varepsilon = 0$)
- Rectangular hyperbola has $a = b$

Mistake 6: Incorrect Parametric Forms

- Parabola: $(at^2, 2at)$
- Ellipse: $(a \cos t, b \sin t)$
- Hyperbola: $(a \sec t, b \tan t)$ or $(a \cosh t, b \sinh t)$

Mistake 7: Forgetting to Shift the Origin

When vertex/center is not at origin:

- Replace x with $(x - h)$
- Replace y with $(y - k)$

Quick Reference: Equations of Tangents

Conic	Tangent at (x_1, y_1)	Tangent at point t
Parabola $y^2 = 4ax$	$yy_1 = 2a(x + x_1)$	$ty - x - at^2 = 0$
Ellipse $x^2/a^2 + y^2/b^2 = 1$	$xx_1/a^2 + yy_1/b^2 = 1$	$x \cos t/a + y \sin t/b = 1$
Hyperbola $x^2/a^2 - y^2/b^2 = 1$	$xx_1/a^2 - yy_1/b^2 = 1$	$x \sec t/a - y \tan t/b = 1$
Rectangular $xy = c^2$	$xy_1 + x_1y = 2c^2$	$x + t^2y = 2ct$

Key Formulas Summary

Parabola ($y^2 = 4ax$)

- Focus: (a, 0)
- Directrix: $x = -a$
- Parametric: $(at^2, 2at)$
- Latus rectum: 4a
- Tangent: $yy_1 = 2a(x + x_1)$

Ellipse ($x^2/a^2 + y^2/b^2 = 1$)

- $b^2 = a^2(1 - \varepsilon^2) = a^2 - c^2$

- Foci: $(\pm c, 0)$ where $c^2 = a^2 - b^2$
- Parametric: $(a \cos t, b \sin t)$
- Latus rectum: $2b^2/a$
- Tangent: $xx_1/a^2 + yy_1/b^2 = 1$

Hyperbola ($x^2/a^2 - y^2/b^2 = 1$)

- $b^2 = a^2(\varepsilon^2 - 1) = c^2 - a^2$
- Foci: $(\pm c, 0)$ where $c^2 = a^2 + b^2$
- Asymptotes: $y = \pm(b/a)x$
- Parametric: $(a \sec t, b \tan t)$
- Latus rectum: $2b^2/a$
- Tangent: $xx_1/a^2 - yy_1/b^2 = 1$

Rectangular Hyperbola ($xy = c^2$)

- Parametric: $(ct, c/t)$
- Tangent: $x + t^2y = 2ct$
- Normal: $y - t^2x + ct^3 - c/t = 0$

Complete MATH 152 Notes — All 5 Parts

Part	Title	Topics Covered
Part 1	Number Systems, Inequalities, Absolute Values, Point Sets	Natural numbers, integers, rational/irrational numbers, inequalities, absolute value, intervals, neighbourhoods, limit points, bounds, Bolzano-Weierstrass theorem
Part 2	Sequences	Arithmetic/geometric sequences, limits of sequences, bounded/monotonic sequences, convergence tests
Part 3	Functions, Limits, Continuity	Function types, bounded/monotonic functions, limits, continuity, differentiability, L'Hospital's rule
Part 4	Rolle's Theorem, Mean Value Theorems, Integration	Rolle's theorem, MVT, Cauchy's MVT, Taylor's theorem, indefinite/definite integration
Part 5	Conic Sections	Parabola, ellipse, hyperbola, equations, tangents, normals, parametric forms